

Retrieval-Augmented Generation (RAG)

1. Why do we need RAG?

LLMs are powerful but have 3 **major limitations**:

1. **Private Data Access** – LLMs can't answer queries about your company/website data (not seen in pre-training).
 - Example: Asking ChatGPT about your course video timestamp → it doesn't know.
2. **Recent Data Access** – LLMs have a **knowledge cut-off date**.
 - Example: Asking "Today's news?" → vanilla LLM fails.
 - Unless connected to live web (like ChatGPT with browsing).
3. **Hallucinations** – LLMs sometimes generate **confident but wrong answers**.
 - Example: "Einstein played football for Germany" → completely false.

👉 RAG was designed to overcome these 3 issues.

2. Alternatives Before RAG

Fine-tuning

- Retrain LLM on your domain-specific dataset.
- Types:
 - **Supervised fine-tuning** → with labeled Q&A pairs.
 - **Continued pretraining** → unlabeled domain corpus.
 - **LoRA / QLoRA / PEFT** → parameter-efficient methods.
 - **RLHF** → reinforcement + human feedback.

✅ Pros: LLM learns your domain deeply.

❌ Cons: Expensive, slow, requires expertise, retraining needed for frequent updates.

In-Context Learning (ICL)

- Provide **examples inside the prompt**.
- Few-shot prompting: model learns from examples without weight updates.
- Example: Sentiment analysis with a few labeled sentences.
- Emergent property of large LLMs (GPT-3+).

✅ Pros: Flexible, no retraining.

❌ Cons: Prompt size limited → can't provide all context.

👉 RAG improves ICL by replacing examples with relevant context chunks.

3. What is RAG?

Definition:

RAG = **Retrieval + Generation**.

- At query time, fetch **relevant external knowledge** (retrieval).
- Inject it into the **prompt** along with user query.
- LLM uses both **parametric knowledge** (inside model) + **contextual knowledge** (retrieved) to generate answer.

Simple analogy:

- LLM = Student's brain (general knowledge).
 - Retrieval = Textbook/notes lookup.
 - RAG = Student uses both memory + textbook to answer exam question.
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4. How does RAG Work? (4 Steps)

Step 1 – Indexing (Build external knowledge base)

1. **Document ingestion** → load PDFs, web pages, videos (LangChain loaders).
 2. **Chunking** → split into smaller, meaningful text chunks (Text Splitters).
 3. **Embedding** → convert chunks into dense vectors using embedding model (e.g., OpenAI embeddings, SBERT).
 4. **Vector storage** → save embeddings + metadata in a vector DB (FAISS, Pinecone, Weaviate, Chroma, Milvus).
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Step 2 – Retrieval (Find relevant context)

- Convert user query → embedding.
 - Search vector DB for top-k closest chunks (semantic similarity).
 - Techniques:
 - **Cosine similarity**
 - **MMR (Maximal Marginal Relevance)**
 - **Contextual compression**
 - **Re-ranking**
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Step 3 – Augmentation (Build prompt)

- Combine **user query** + **retrieved chunks** into a structured prompt.
- Example:

You are a helpful assistant.

Answer only from the provided context.

If the context is insufficient, say "I don't know".

Context: [retrieved text chunks]

Question: [user query]

Step 4 – Generation (Answer using LLM)

- LLM processes the augmented prompt.
 - Uses both **its parameters** + **injected context**.
 - Generates **grounded, accurate response**.
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5. How RAG Solves the 3 Problems

1. **Private Data** → Context comes from your company's docs, so model can answer.
 2. **Recent Data** → Just keep updating vector DB with new info (no retraining).
 3. **Hallucination** → Prompt explicitly restricts model to provided context.
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6. Advantages of RAG over Fine-tuning

- **Cheaper** (no retraining needed).
 - **Faster** (just add new docs to DB).
 - **Flexible** (works with multiple sources).
 - **Safer** (reduces hallucination by grounding answers).
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7. Limitations of RAG

- Retrieval quality depends on:
 - Good **chunking strategy**
 - Good **embeddings**
 - Good **vector DB search**

- Large knowledge bases may need **optimization** (re-ranking, hybrid search).
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8. Interview-Style Q&A

Q1: Why do we need RAG if we already have fine-tuning?

- Fine-tuning is costly, slow, not scalable for dynamic data.
- RAG is lightweight, cheaper, supports real-time updates.

Q2: What are the 4 main steps of RAG?

- Indexing, Retrieval, Augmentation, Generation.

Q3: How does RAG reduce hallucinations?

- By grounding answers in retrieved context + instructing model to only use it.

Q4: Which vector databases are commonly used?

- Local: FAISS, Chroma.
- Cloud: Pinecone, Weaviate, Milvus, Qdrant.

Q5: When would you still prefer fine-tuning over RAG?

- When model must deeply internalize domain-specific reasoning (e.g., medical diagnosis, legal).

Q6: How does RAG handle recent information?

- Add new documents to DB → embeddings → stored. No retraining.
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9. Real-World Applications of RAG

- **Enterprise Chatbots** → Answer from internal documents.
- **E-learning Assistants** → Student queries from lecture transcripts.
- **Legal/Medical Search** → Summarize domain documents.
- **Customer Support** → FAQs + manuals retrieval.